



ROOT CAUSE UNLOCKED

The Complete Patient's Guide to *Tarsal* Tunnel Syndrome

Natural Strategies for Foot Nerve Pain Relief, Healing, and Recovery

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Complete Patient's Guide to Tarsal Tunnel Syndrome

Natural Strategies for Foot Nerve Pain Relief, Healing, and Recovery

BEFORE YOU START

Disclaimer: This ebook is for educational purposes only and does not constitute medical advice. Always consult with a qualified healthcare provider before starting any new treatment, supplement, exercise program, or dietary change. Never start, stop, or change a prescribed medication without talking to the prescribing clinician first. If you experience progressive weakness, severe swelling with fever, loss of sensation spreading upward, or symptoms that worsen rapidly, seek medical attention immediately.

A note about the evidence in this guide. Every study referenced in this edition has been checked against the published medical literature and carries a PubMed identifier (PMID) so you can look it up yourself. Where a treatment is widely used but has not been studied specifically in tarsal tunnel syndrome, this guide says so plainly rather than implying evidence that does not exist. Where a study was done in a different population (for example, plantar fasciitis or diabetic neuropathy), that scope is stated in the text. Nothing here promises a cure or a guaranteed result.

Contents

1. What Is Tarsal Tunnel Syndrome?

2. Understanding the Anatomy

3. Common Causes and Risk Factors

4. How Tarsal Tunnel Syndrome Is Diagnosed

5. The Healing Timeline: What to Expect

6. Supplements Discussed for Nerve Health and Inflammation

7. Targeted Exercises for Tarsal Tunnel Relief

8. Physical Therapies and Procedures

9. The Anti-Inflammatory Diet for Nerve Recovery

10. Lifestyle Modifications and Pain Management

11. The In-Clinic Pre-Treatment Protocol

12. Red Flags: When to Seek Immediate Care

13. Building Your Personal Recovery Plan

Final Thoughts

References

Working with Root Health

1. What Is Tarsal Tunnel Syndrome?

Tarsal Tunnel Syndrome (TTS) is a compression neuropathy affecting the tibial nerve as it travels through the tarsal tunnel, a narrow passageway on the inside (medial aspect) of your ankle. It is often described as the “carpal tunnel syndrome of the foot,” though it is far less common than its wrist counterpart.^{1,2}

The condition occurs when the tibial nerve or its branches (the medial plantar nerve, lateral plantar nerve, and medial calcaneal nerve) become compressed, irritated, or entrapped within the tarsal tunnel, leading to pain, numbness, tingling, and burning sensations in the foot.^{1,2}

Common Symptoms

- Burning, tingling, or “pins and needles” on the sole of the foot (especially the heel, arch, and toes)
- Numbness in the bottom of the foot that may worsen with standing or walking
- Shooting or electric pain radiating from the inside of the ankle into the foot
- Pain that worsens with activity such as prolonged standing, walking, or running
- Pain that improves with rest and leg elevation
- Nighttime symptoms: burning or tingling that wakes you from sleep
- Weakness in the intrinsic foot muscles (toe spreading, foot arch control)
- Positive Tinel’s sign: tapping over the tarsal tunnel reproduces symptoms

Important: Unlike plantar fasciitis (which causes heel pain primarily with the first steps in the morning), TTS symptoms often worsen as the day progresses and with weight-bearing activity.

2. Understanding the Anatomy

The tarsal tunnel is a fibro-osseous canal located on the medial (inner) side of the ankle, posterior to the medial malleolus (the prominent bony bump on the inside of your ankle).³

Boundaries of the Tarsal Tunnel

The roof of the tarsal tunnel is formed by the flexor retinaculum, a thick band of connective tissue also called the lacinate ligament. Its floor is the medial surfaces of the talus and calcaneus bones. Within the tunnel run the tibial nerve, the posterior tibial artery and veins, and four tendons: the tibialis posterior, the flexor digitorum longus, the flexor hallucis longus, and their tendon sheaths.

The Tibial Nerve Branches

As the tibial nerve exits the tarsal tunnel, it divides into three main branches. The medial plantar nerve supplies sensation to the medial sole, the big toe, and the second, third, and fourth toes, and also innervates the abductor hallucis, flexor digitorum brevis, flexor hallucis brevis, and the first lumbrical muscles. The lateral plantar nerve supplies sensation to the lateral sole and the little toe, and innervates most of the intrinsic foot muscles, including the quadratus plantae, abductor digiti minimi, and the interossei. The medial calcaneal nerve provides sensation to the medial and central heel.

Because the tarsal tunnel is a fixed, non-expandable space, any structure that takes up extra room (swelling, varicose veins, ganglion cysts, bone spurs, or inflamed tendon sheaths) can compress the tibial nerve and trigger symptoms.³

3. Common Causes and Risk Factors

A systematic review of the tarsal tunnel literature reports that the syndrome is most commonly idiopathic, meaning no single cause is found. Other reported causes include prior trauma, lipomas, cysts, ganglia, schwannomas, varicose plantar veins, anatomic anomalies, and systemic inflammatory conditions.⁴

Primary Causes

In many cases no single cause is identified: the nerve becomes compressed within the tight confines of the tarsal tunnel, often attributed to repetitive microtrauma, an idiopathic pattern of intrinsic compression.⁴ Flat feet, or pes planus, and overpronation are commonly cited contributing factors, since excessive inward rolling of the foot changes the geometry of the tarsal tunnel and increases tension on the tibial nerve. Varicose veins within the tunnel can occupy space and compress the nerve,⁴ and ganglion cysts or lipomas are space-occupying lesions that do the same.⁴ Bone spurs or fracture malunions, bony growths or misaligned bone fragments left over from old ankle or calcaneal fractures, can narrow the tunnel. Tenosynovitis, inflammation and thickening of the tendon sheaths (especially the flexor tendons) within the tunnel, reduces the space available to the nerve. Systemic conditions reported among the associated causes include diabetes, hypothyroidism, rheumatoid arthritis, systemic sclerosis, chronic kidney disease, and amyloidosis.⁴ Ankle sprains or other trauma can predispose to TTS through swelling, scar tissue, or altered biomechanics after the injury.⁴ Posterior tibial tendon dysfunction, a problem with this key arch-supporting tendon, alters foot mechanics and increases strain on the tarsal tunnel contents.

Risk Factors

Reported risk factors include female sex, athletic participation, hypothyroidism, diabetes mellitus, systemic sclerosis, chronic renal failure, and hemodialysis use.⁴ Tarsal tunnel syndrome is also described in runners as a cause of foot pain that can be mistaken for other overuse conditions.⁵

Additional factors commonly discussed in clinical practice include age, since the condition is most often described in adults in mid-life, and higher body weight, which increases mechanical load on the feet and can contribute to venous congestion. Occupations requiring prolonged standing, such as nursing, teaching, retail work, and factory work, are frequently mentioned, as are athletic pursuits like running (especially with overpronation), dancing, and hiking. Pregnancy is a contributing factor because fluid retention and weight gain increase tunnel pressure, and a history of previous ankle surgery or trauma can leave scar tissue and altered anatomy that predispose to the condition.

4. How Tarsal Tunnel Syndrome Is Diagnosed

TTS is primarily a clinical diagnosis, but objective testing is often used to support the diagnosis and rule out other conditions.

Clinical Examination

Palpation: Tenderness over the tarsal tunnel (posterior to the medial malleolus). Your provider may feel for masses, swelling, or thickened tendon sheaths.

Special Tests: Your provider may use several maneuvers to help confirm the diagnosis. Tinel's sign involves tapping over the tarsal tunnel; reproducing tingling, burning, or electric sensations radiating into the foot is a positive test that suggests tibial nerve irritation. In the dorsiflexion-eversion test, the examiner dorsiflexes (pulls up) and everts (turns outward) the ankle while extending the toes, and holds the position. This maneuver was described specifically for tarsal tunnel syndrome and reproduced or intensified symptoms in the patients studied,⁶ and a later biomechanical study confirmed that this position increases strain on the tibial nerve and its plantar branches, which is the mechanism the test relies on.⁷ The plantar flexion-inversion test, in which the ankle is plantar flexed (pointing down) and inverted (turning inward) and held in that position, can also reproduce symptoms in some patients. Two-point discrimination testing checks your reduced ability to distinguish between two closely spaced points on the sole of the foot, which suggests sensory nerve involvement, and muscle strength testing looks for weakness in toe flexion, toe spreading, or foot inversion that may indicate motor nerve involvement.

Imaging and Diagnostic Tests

MRI: Widely used to visualize the tarsal tunnel contents. It can show space-occupying lesions (cysts, lipomas, varicose veins), tendon pathology, bone changes, and nerve swelling. MRI has been used in the evaluation of tarsal tunnel syndrome since the early 1990s.⁸

Ultrasound: Increasingly used because it is practical, inexpensive, and allows dynamic assessment. In a surgical series of 17 feet, preoperative ultrasound identified space-occupying lesions (ganglia, talocalcaneal coalition, varicose veins) that were then confirmed at surgery, including three ganglia that could not be felt on examination. The authors concluded that ultrasound is useful for identifying space-occupying lesions in suspected tarsal tunnel syndrome.⁹ A large illustrated review catalogs the range of causes that ultrasound can demonstrate, including retinacular thickening, tenosynovitis, accessory muscles, cysts, tumors, and vascular abnormalities.¹⁰ Ultrasound is less expensive than MRI and can guide injections.

Nerve Conduction Studies (NCS) and Electromyography (EMG): These measure the speed and strength of electrical signals in the tibial nerve and its branches. An evidence-based review by the American Association of Neuromuscular and Electrodiagnostic Medicine examined how useful these techniques are in suspected tarsal tunnel syndrome and is the standard reference clinicians use when deciding which

electrodiagnostic studies to order.¹¹ Results should always be interpreted alongside your history and examination, not in isolation.

X-rays: Limited utility for soft tissues, but helpful for ruling out bony abnormalities, arthritis, or fracture malunions.

Differential Diagnosis

Your healthcare provider will also consider:

- **Plantar Fasciitis:** Heel pain, typically worse with the first steps in the morning and localized to the medial heel
- **Peripheral Neuropathy (for example diabetic or alcohol-related):** Usually bilateral, symmetric, “stocking-glove” distribution
- **Lumbar Radiculopathy (S1 nerve root):** Can cause similar foot symptoms, but usually with back pain and a positive straight leg raise
- **Morton’s Neuroma:** Pain between the toes (typically the 3rd to 4th web space), not the sole
- **Peripheral Artery Disease:** Causes claudication pain rather than tingling and burning
- **Calcaneal Stress Fracture:** Localized heel pain with weight-bearing
- **Complex Regional Pain Syndrome (CRPS):** Severe, disproportionate pain with color and temperature changes

5. The Healing Timeline: What to Expect

Understanding the usual course of TTS helps set realistic expectations and guides treatment decisions.

Natural History

In the acute phase, from zero to six weeks, symptoms are most noticeable and rest, activity modification, and conservative measures are started. The subacute phase, from six weeks to six months, is the working window for structured rehabilitation, though symptoms may fluctuate during this time. If symptoms persist beyond six months into the chronic phase, more intensive or interventional treatments may be considered.

What the Outcome Literature Actually Says

A systematic review comparing conservative and surgical management concluded that conservative management remains a good option that can resolve symptoms for many patients, and that surgery is currently pursued in a select group of patients whose symptoms do not respond to medical or minimally invasive treatment. The same review found that surgical outcomes reported in the literature are generally good, and that the evidence base for surgery is stronger than that for minimally invasive alternatives.⁴

Beyond that review, the surgical evidence consists largely of case series from single centers, each with different patient selection, different definitions of success, and different follow-up lengths.^{12,13,14,15} That is why published success rates vary so widely between reports, and it is why no single percentage should be treated as your personal odds. Ask your surgeon about their own results in patients like you.

Recurrence: Symptoms can return, particularly if underlying biomechanical issues such as overpronation are not addressed. Recurrent tarsal tunnel syndrome after surgery is a recognized clinical problem with its own management considerations.¹⁶

Key insight: Identifying and addressing the underlying cause matters. When imaging finds a specific space-occupying lesion such as a ganglion or a coalition, there is a definable target to treat.^{9,10} When no cause is found, treatment focuses on unloading the nerve and correcting mechanics.

6. Supplements Discussed for Nerve Health and Inflammation

The supplements below are commonly discussed for nerve health. Read this section carefully, because the honest summary is this: **almost none of these have been studied in tarsal tunnel syndrome specifically.** Most of the research was done in diabetic peripheral neuropathy, in joint arthritis, or in general pain populations. That does not make them useless, but it does mean the evidence is indirect. Tarsal tunnel syndrome is fundamentally a mechanical compression problem, and no supplement decompresses a nerve.

Always discuss supplements with your healthcare provider, especially if you take medications. Never stop or change a prescribed medication on your own.

6.1 Methylcobalamin (Active Vitamin B12)

What it does: Vitamin B12 is required for myelin (the protective sheath around nerves). B12 deficiency is a well-recognized cause of peripheral neuropathy, which is why clinicians often check B12 levels in patients with unexplained nerve symptoms.

Evidence: A systematic review of clinical controlled trials examined the effectiveness of vitamin B12 in **diabetic** neuropathy.¹⁷ This is not a study of tarsal tunnel syndrome. B12 is most clearly worth addressing when a deficiency is actually documented on blood testing.

Typical dosage discussed: 1,000 to 2,000 mcg daily (sublingual or oral). **Caution:** Generally well tolerated. Ask your provider whether testing your level makes sense before supplementing.

6.2 Alpha-Lipoic Acid (ALA)

What it does: An antioxidant that has been studied for its effects on oxidative stress and microcirculation in nerve tissue.

Evidence: Alpha-lipoic acid has been studied more extensively than most supplements in this list, but the research population is **diabetic polyneuropathy**, not nerve compression. The literature includes a meta-analysis of short-term treatment in symptomatic diabetic polyneuropathy,¹⁸ a two-year multicenter randomized double-blind placebo-controlled trial (ALADIN II),¹⁹ and a four-year randomized trial (NATHAN 1).²⁰ None of these enrolled patients with tarsal tunnel syndrome.

Typical dosage discussed: 300 to 600 mg daily. **Caution:** May lower blood sugar, which matters if you take glucose-lowering medication. Discuss with your prescriber.

6.3 Acetyl-L-Carnitine (ALCAR)

What it does: Involved in mitochondrial energy metabolism and studied for effects on nerve regeneration.

Evidence: A pooled analysis of two randomized placebo-controlled trials reported improvements in pain, nerve regeneration, and vibratory perception in patients with chronic **diabetic** neuropathy.²¹ An earlier open study examined L-acetylcarnitine in painful peripheral neuropathies of mixed cause.²² Again, neither addressed tarsal tunnel syndrome.

Typical dosage discussed: 500 to 1,000 mg twice daily. **Caution:** May cause insomnia if taken late in the day. Discuss with your provider if you have a seizure disorder.

6.4 Omega-3 Fatty Acids (EPA/DHA)

What it does: EPA and DHA are incorporated into cell membranes and are precursors to mediators involved in resolving inflammation.

Evidence: A review of omega-3 fatty acids and inflammatory processes summarizes the biology, from molecular mechanisms through human effects.²³ This is background biology, not a clinical trial in foot nerve pain. There is no trial of omega-3 supplementation in tarsal tunnel syndrome.

Typical dosage discussed: 1,200 to 2,400 mg combined EPA/DHA daily. Triglyceride or re-esterified triglyceride forms are often preferred for absorption. That is the range used in the one study run in people with nonsurgical neck and back pain, where most participants took 1,200 mg daily and about a fifth took 2,400 mg. That study was an uncontrolled questionnaire follow-up rather than a placebo-controlled trial, so treat it as a starting point rather than a proven dose. Higher intakes appear in the joint-pain literature, but that work was done in rheumatoid arthritis and other inflammatory conditions rather than in the mechanical and degenerative problems this guide covers, so I have not carried its dose across.⁴⁶ **Caution:** May increase bleeding risk at higher doses. Tell your surgeon if you are having a procedure, and follow their instructions.

6.5 Palmitoylethanolamide (PEA)

What it does: A naturally occurring fatty acid amide studied for anti-inflammatory and neuroprotective effects, particularly through actions on mast cells and glial cells.

Evidence: A pooled data meta-analysis examined PEA as a food for special medical purposes in the treatment of chronic pain.²⁴ There is also a published case series describing PEA in neuropathic pain from a variety of causes; case series are a low level of evidence and cannot establish effectiveness.²⁵ Neither source studied tarsal tunnel syndrome.

Typical dosage discussed: 300 to 600 mg twice daily (micronized form is the form usually studied). The upper end of that range is the dose used in a randomized trial of ultra-micronized PEA at 600 mg twice daily in older adults with chronic pain.⁴⁴ **Caution:** Generally well tolerated in the published reports.

6.6 Benfotiamine (a Fat-Soluble Vitamin B1 Derivative)

What it does: A lipid-soluble derivative of thiamine that is better absorbed than water-soluble thiamine and is discussed in relation to advanced glycation end-product formation.

Evidence: A benfotiamine and B-vitamin combination was studied in **diabetic** polyneuropathy,²⁶ and a separate study compared different benfotiamine dosage regimens in painful **diabetic** neuropathy.²⁷ Both are older studies in a diabetic population, not in nerve compression.

Typical dosage discussed: 150 to 300 mg twice daily. That range corresponds to the two active arms of the BENDIP trial, which compared benfotiamine at 300 mg and 600 mg per day against placebo in diabetic polyneuropathy.⁴⁵ **Caution:** Generally well tolerated.

6.7 Turmeric / Curcumin

What it does: Curcumin inhibits several inflammatory signaling pathways in laboratory studies.

Evidence: The human clinical evidence for curcumin is in **joint arthritis**, not nerve compression. A systematic review and meta-analysis of randomized trials evaluated turmeric extracts and curcumin for arthritis symptoms.²⁸ A multicenter trial compared a Curcuma domestica extract with ibuprofen in patients with **knee osteoarthritis**.²⁹ Applying these results to tarsal tunnel syndrome is an extrapolation, not a finding.

Typical dosage discussed: 500 to 1,000 mg curcuminoids daily, often with piperine or in a liposomal or phytosome form for absorption. **Caution:** May interact with blood thinners. Discuss with your prescriber. Often avoided in active gallbladder disease.

6.8 Magnesium

What it does: Involved in muscle relaxation, NMDA receptor modulation, and normal nerve conduction.

Evidence: The published supplementation trial most often cited here studied magnesium and L-carnitine in **migraine prophylaxis**.³⁰ That is a different condition. There is no evidence that magnesium supplementation treats tarsal tunnel syndrome. Some people use magnesium at night for general muscle relaxation and sleep, which is a comfort measure rather than a treatment for nerve compression.

Typical dosage discussed: 200 to 400 mg elemental magnesium daily. Glycinate is a commonly chosen form for tolerability. This is a general supplemental intake range rather than a dose drawn from a trial in this condition, which is why no study is cited beside it. **Caution:** Can cause loose stools. Avoid in severe kidney disease unless your provider directs otherwise.

6.9 Vitamin D

What it does: Involved in muscle function, immune regulation, and bone health. Deficiency is common.

Evidence: A frequently cited study found a high prevalence of severe vitamin D deficiency among patients presenting with persistent, nonspecific **musculoskeletal** pain.³¹ Note carefully what that study is: it measured how often deficiency was present, not whether supplementing fixed the pain. It does not show that vitamin D treats neuropathic pain or tarsal tunnel syndrome. The practical takeaway is that it is reasonable to have your level checked if you have persistent pain, and to correct a documented deficiency.

Typical dosage discussed: Dosing should be guided by your blood level and your provider's advice. D3 (cholecalciferol) is the usual form. **Caution:** Have levels checked rather than guessing at a dose.

6.10 N-Acetylcysteine (NAC)

What it does: A precursor to glutathione, the body's main intracellular antioxidant.

Evidence: Published reviews describe NAC's antioxidant activity, its effects on inflammatory signaling, and its established clinical uses.³² There is no clinical trial of NAC in tarsal tunnel syndrome, and the general antioxidant rationale should not be read as evidence of benefit for compressed nerves.

Typical dosage discussed: 600 to 1,200 mg daily. **Caution:** Can cause nausea. Discuss with your provider if you take nitroglycerin.

6.11 Coenzyme Q10 (CoQ10)

What it does: A component of mitochondrial energy production that also functions as an antioxidant.

Evidence: A review covers CoQ10 supplementation in aging and disease broadly.³³ It is background science, not evidence for tarsal tunnel syndrome.

Typical dosage discussed: 100 to 200 mg daily. Ubiquinol is often chosen for absorption. **Caution:** May interact with blood thinners. Discuss with your prescriber.

A Realistic Summary of This Chapter

If you take nothing else from Chapter 6, take this: correcting a documented deficiency (B12, vitamin D) is sensible medicine. Everything else in this list is optional, indirectly evidenced, and secondary to the things that actually change the mechanics of your foot. Spend your effort and money on footwear, load management, and rehabilitation first.

7. Targeted Exercises for Tarsal Tunnel Relief

Critical rule: Stop any exercise that increases burning, tingling, or numbness in the foot. A mild stretching sensation is acceptable. Increased neuropathic symptoms mean the nerve is being irritated. If in doubt, ask your physical therapist or physician before continuing.

7.1 Tibial Nerve Glides (Neural Flossing)

Best for: Improving tibial nerve mobility and tolerance to movement

How to perform (seated):

1. Sit in a chair with your affected leg extended straight out in front of you
2. Slowly dorsiflex your ankle (pull toes toward your shin) while extending your knee
3. At the same time, gently flex your neck forward (tuck chin toward chest)
4. Then slowly plantarflex your ankle (point toes away) while slightly bending your knee
5. At the same time, gently return your neck to neutral
6. This creates a “flossing” motion of the nerve
7. Perform 10 slow, controlled repetitions
8. Never push into pain. Mild tingling may be acceptable, but sharp pain or increased numbness is not

Why it works: Neurodynamic techniques are usually divided into “sliders,” which move the nerve relative to surrounding tissue with minimal increase in nerve tension, and “tensioners,” which deliberately increase tension. The distinction matters clinically, and the sliding technique described above is the gentler of the two, which is why it is the appropriate starting point for an irritable nerve.³⁴ A systematic review of randomized controlled trials of neural mobilization analyzed the therapeutic efficacy of these techniques and noted that the evidence base at the time was limited and the trials were heterogeneous.³⁵ In other words, this is a reasonable, low-risk technique with modest supporting evidence, not a proven cure.

7.2 Intrinsic Foot Strengthening

Best for: Supporting the arch and improving foot mechanics

Short Foot Exercise (doming):

1. Stand or sit with your foot flat on the floor
2. Without curling your toes, try to “shorten” your foot by drawing the ball of your foot toward your heel
3. You should see your arch rise slightly
4. Hold for 5 seconds, repeat 10 to 15 times

5. Perform 2 to 3 sets daily

Toe Spreading:

1. Sit with your foot flat
2. Spread your toes apart as wide as possible
3. Hold for 5 seconds, repeat 10 to 15 times

Towel Scrunches:

1. Place a small towel on the floor
2. Use your toes to scrunch the towel toward you
3. Continue for 1 to 2 minutes

Why it works: The “foot core” model describes the intrinsic foot muscles as an active subsystem that supports the medial longitudinal arch alongside passive (ligamentous) and neural subsystems, and argues that these muscles are trainable and clinically relevant.³⁶ A study of plantar intrinsic muscle training examined its effect on medial longitudinal arch morphology and dynamic function.³⁷ These studies address arch mechanics, not tarsal tunnel syndrome outcomes, but arch mechanics are one of the levers you can actually control.

7.3 Calf Stretching

Best for: Improving ankle dorsiflexion and reducing tension through the posterior leg

Gastrocnemius Stretch (straight knee):

1. Stand facing a wall, hands on the wall at shoulder height
2. Step the affected leg back, keeping the knee straight and heel flat on the floor
3. Lean forward until you feel a stretch in the upper calf
4. Hold for 30 seconds, repeat 3 times

Soleus Stretch (bent knee):

1. Same position as above, but bend the back knee slightly
2. You should feel the stretch lower in the calf, near the Achilles
3. Hold for 30 seconds, repeat 3 times

Why it works: Tight calf muscles restrict ankle dorsiflexion, which alters how the foot loads during walking and running. This is a mechanical rationale rather than a treatment proven in tarsal tunnel syndrome, but restoring normal ankle motion is a standard part of foot and ankle rehabilitation.

7.4 Ankle Mobility and Dorsiflexion

Best for: Restoring normal ankle range of motion

Ankle Circles:

1. Sit or lie down with your leg extended
2. Slowly rotate your ankle in large circles, 10 clockwise and 10 counterclockwise
3. Perform 2 to 3 times daily

Knee-to-Wall Dorsiflexion:

1. Stand facing a wall with your toes 2 to 4 inches from the wall
2. Keeping your heel flat, try to touch your knee to the wall
3. If successful, move your foot back slightly and repeat
4. Find the maximum distance where you can still touch the wall with your heel down
5. Hold for 5 seconds, repeat 10 times

Why it works: Limited ankle dorsiflexion tends to be compensated for elsewhere in the foot, often with more pronation or an earlier heel lift. Restoring motion removes that compensation.

7.5 Posterior Tibialis Strengthening

Best for: Supporting the arch and controlling overpronation

Resisted Inversion with Band:

1. Sit with your leg extended, loop a resistance band around the forefoot
2. Anchor the other end to a stable object on the outside of your foot
3. Slowly turn your foot inward (inversion) against the band's resistance
4. Hold for 2 seconds, return slowly
5. Perform 2 to 3 sets of 12 to 15 repetitions

Heel Raises with Inversion Bias:

1. Stand with your feet slightly turned inward
2. Slowly rise up onto your toes
3. Hold for 2 seconds, lower slowly
4. Perform 2 to 3 sets of 10 to 15 repetitions

Why it works: The posterior tibialis tendon passes through the tarsal tunnel and is the primary dynamic supporter of the medial arch. Strengthening it is a standard component of arch and pronation management.

7.6 Balance and Proprioception

Best for: Improving foot stability and neuromuscular control

Single-Leg Stand:

1. Stand on your affected leg near a wall or counter for safety
2. Hold for 30 seconds, gradually increasing to 60 seconds
3. Progress by closing your eyes or standing on a cushion

Toe Walking: Walk on your toes for 20 to 30 meters, rest, then repeat.

Heel Walking: Walk on your heels for 20 to 30 meters, rest, then repeat.

Why it works: Balance training recruits the small stabilizing muscles of the foot and ankle and is a routine part of lower limb rehabilitation.

8. Physical Therapies and Procedures

8.1 Physical Therapy (PT)

Physical therapy is the cornerstone of conservative TTS management. A skilled physical therapist will assess your gait, foot posture, and biomechanics, prescribe nerve gliding exercises and progressive strengthening, and use manual therapy to improve ankle and foot joint mobility. They will also address contributing factors such as tight calves, weak intrinsic foot muscles, and overpronation, educate you on footwear and orthoses, and progress you through a structured return-to-activity program.

Evidence: A systematic review comparing conservative and surgical management of tarsal tunnel syndrome concluded that conservative management remains a good first-line option that can resolve symptoms for many patients.⁴ The review did not isolate physical therapy from the rest of conservative care, so treat this as support for the overall conservative approach rather than for any single technique.

8.2 Orthoses and Footwear Modification

Proper footwear and foot orthoses are among the most practical interventions available to you, particularly when overpronation appears to be contributing. Orthoses generally aim to provide medial arch support that reduces overpronation, heel cups or wedges that influence rearfoot alignment, and metatarsal pads that redistribute forefoot pressure.

Evidence, stated honestly: There is no randomized trial of custom orthoses in tarsal tunnel syndrome. The orthosis evidence that does exist comes from other foot conditions. A randomized trial of foot orthoses in **plantar fasciitis** found short-term benefits for pain and function, with the effects diminishing over longer follow-up.³⁸ A randomized trial in people with **excessively pronated feet** examined the effect of rigid foot orthoses on balance parameters.³⁹ These studies support the general principle that orthoses can change foot mechanics and can help some foot conditions. They do not demonstrate that orthoses treat tarsal tunnel syndrome, and this guide does not claim they do.

Footwear recommendations: Look for shoes with good arch support and a firm heel counter, and avoid flat, unsupportive shoes such as flip-flops, ballet flats, or worn-out sneakers. If you overpronate, running shoes with motion control or stability features are worth considering. Replace running shoes when the midsole feels flat or the wear pattern becomes obvious; common practical guidance puts this somewhere in the 300 to 500 mile range, but that is a rule of thumb, not a research finding.

8.3 Night Splints

A night splint holds the ankle in a neutral position (slight dorsiflexion) during sleep, on the reasoning that a plantarflexed foot position may increase tension in the region overnight.

Evidence, stated honestly: This guide could not identify a published study of night splints in tarsal tunnel syndrome. The rationale is mechanical and is borrowed from the use of night splints in other foot and nerve compression conditions. If you try one, treat it as a low-cost experiment: if morning symptoms improve over a few weeks, continue; if not, stop. Some patients find a dorsal night splint more comfortable than a boot-style splint.

8.4 Manual Therapy

Manual therapy techniques address soft tissue restrictions, joint mobility, and nerve mobility. Techniques your therapist may use include soft tissue mobilization of the calf and foot intrinsic muscles, ankle joint mobilization to improve dorsiflexion, gentle techniques directed at the tarsal tunnel region, and myofascial techniques for the plantar fascia and foot muscles.

Evidence, stated honestly: Manual therapy is standard practice within physical therapy, but this guide could not identify a study testing manual therapy specifically in tarsal tunnel syndrome. It is described here as a component of care your therapist may use, not as an independently proven treatment. The neurodynamic component is discussed with its evidence in section 7.1.^{34,35}

8.5 Taping

Low-Dye Taping: A taping technique applied to support the medial longitudinal arch and limit excessive foot motion.

How it works: Athletic tape is applied in a specific pattern to lift the arch and control foot motion. It is often used as a short trial to see whether arch support helps before investing in custom orthoses.

Evidence: A randomized trial found that low-Dye taping produced a small improvement in “first-step” pain over one week in people with **plantar heel pain**.⁴⁰ That is a different condition and a short time frame, so read this as evidence that the technique changes symptoms in a pronation-related foot problem, not as evidence for tarsal tunnel syndrome.

8.6 Acupuncture and Dry Needling

Acupuncture involves inserting thin needles at specific points. Dry needling targets myofascial trigger points in the calf and foot muscles.

Evidence: An individual patient data meta-analysis of acupuncture trials in **chronic pain** (including back and neck pain, osteoarthritis, chronic headache, and shoulder pain) found effects that were statistically significant compared with both sham and no-acupuncture controls, though the difference from sham was modest.⁴¹ Tarsal tunnel syndrome was not among the conditions studied. If you pursue acupuncture, frame it as a general pain-coping option rather than a treatment for the compression itself.

8.7 Extracorporeal Shockwave Therapy (ESWT)

ESWT delivers acoustic pulses to tissue and is used in some tendon conditions.

Evidence, stated honestly: I searched the published literature and could not identify any study of extracorporeal shockwave therapy in tarsal tunnel syndrome specifically. ESWT is in this guide only so you recognize the term if it is offered to you, and so you know to ask what evidence supports it for your diagnosis rather than for a different tendon problem.

8.8 Corticosteroid Injections

Corticosteroids are sometimes injected into or around the tarsal tunnel with the aim of reducing local inflammation and swelling, particularly when there is tenosynovitis. Injections can also serve a diagnostic purpose: a temporary response localized to the tunnel supports the diagnosis.

Evidence, stated honestly: I could not verify a published trial of corticosteroid injection specifically in tarsal tunnel syndrome. What is behind this section is clinical practice rationale, not trial evidence, and you should price the decision accordingly.

The practical view: An injection may offer temporary relief and may help clarify the diagnosis, but it does not remove the cause of compression. Repeated injections carry recognized risks including nerve injury, tendon weakening, and fat pad atrophy. This is a decision to make with your physician, who can weigh your specific anatomy and imaging.

8.9 Surgery (Tarsal Tunnel Release)

Surgery involves releasing the flexor retinaculum to decompress the tibial nerve, and addressing any space-occupying lesion found. It is generally considered when conservative treatment has failed over a sustained period, when there is a confirmed space-occupying lesion such as a cyst, coalition, varicosity, or tumor, when there is progressive motor weakness or muscle atrophy, or when symptoms are severely disabling.

Evidence, stated honestly: A systematic review concluded that surgery is currently reserved for a select group of patients with refractory symptoms, that reported surgical outcomes in the literature are generally good, and that the evidence for surgery is stronger than for minimally invasive alternatives.⁴ The underlying primary literature is mostly case series from individual centers with varying patient selection and varying definitions of success.^{12,13,14,15} You will see specific success percentages quoted for this operation in a lot of places. I am not going to repeat one, because I could not verify a source that supports a single number, and the honest answer is that success here depends heavily on whether the diagnosis was right in the first place. Ask your surgeon what their own results look like and how they select patients.

Considerations: Surgery helps many patients but not all. Risks include infection, incomplete relief, scar tissue formation, recurrent symptoms,¹⁶ and complex regional pain syndrome. Discuss the realistic range of outcomes with your surgeon, in your own case, before deciding.

9. The Anti-Inflammatory Diet for Nerve Recovery

Nutrition supports general tissue health and overall wellbeing. To be clear about what this chapter is: it is sensible general eating guidance, not a treatment for nerve compression. No diet decompresses a nerve.

9.1 Foods to Emphasize

For nerve-supporting nutrients, look to B-vitamin rich foods such as beef, eggs, fish, nutritional yeast, and leafy greens for B12, B1, B6, and folate; omega-3 rich foods such as salmon, sardines, mackerel, walnuts, flaxseeds, and chia seeds; antioxidant-rich foods such as berries, dark leafy greens, and colorful vegetables; and sulfur-rich foods such as garlic, onions, eggs, and cruciferous vegetables.

Anti-inflammatory foods worth building meals around include leafy greens like spinach, kale, and Swiss chard; berries such as blueberries, strawberries, and blackberries; cruciferous vegetables like broccoli, cauliflower, and Brussels sprouts; turmeric with black pepper; ginger; and green tea.

For quality protein, favor meats, poultry, fish, eggs, and legumes if tolerated, and for healthy fats, favor extra virgin olive oil, avocados, and nuts and seeds.

9.2 Foods to Minimize

Pro-inflammatory fats are worth limiting: trans fats from partially hydrogenated oils, heavily refined vegetable oils, and deep-fried foods. Refined carbohydrates and sugars, including white bread, pastries, sugary cereals, soda and sweetened beverages, and candy and desserts, are another category to minimize. Diets high in refined sugar are associated with poorer metabolic health, which matters especially if you have diabetes or prediabetes, since diabetes is itself a risk factor for nerve problems.⁴ Processed foods such as deli meats, hot dogs, bacon, fast food, and packaged snacks with artificial ingredients are best limited as well. Excessive alcohol deserves particular attention, since alcohol excess is a recognized cause of peripheral neuropathy in its own right and heavy drinking also interferes with sleep. High-sodium foods may contribute to fluid retention and swelling.

9.3 Hydration

Adequate hydration supports general physiology. A commonly used practical target is roughly half your body weight in pounds, taken as ounces of water daily. For a 160-pound person, that is about 80 ounces (roughly 10 cups). This is a rule of thumb, not a research-derived prescription, and your needs vary with activity, climate, and medical conditions. If you have heart or kidney disease, follow your provider's fluid guidance instead.

A few practical habits help: start your day with 16 to 20 oz of water, drink before meals, limit caffeine and alcohol, and add electrolytes if you sweat heavily.

9.4 Sample Anti-Inflammatory Day

Breakfast: Smoothie with spinach, blueberries, flaxseed, ginger, almond butter, and unsweetened almond milk. Or oatmeal with walnuts, chia seeds, cinnamon, and berries.

Mid-morning snack: A handful of walnuts and an apple. Or hard-boiled eggs.

Lunch: Large salad with mixed greens, grilled salmon, avocado, cherry tomatoes, olive oil and lemon dressing. Side of vegetable soup.

Afternoon snack: Carrot and celery sticks with hummus. Or pumpkin seeds.

Dinner: Beef or chicken with turmeric and garlic. Steamed broccoli and sweet potato. Side salad with olive oil.

Evening: Ginger or turmeric tea. A small square of dark chocolate (70% or higher cacao).

10. Lifestyle Modifications and Pain Management

10.1 Activity Modification

The most important lifestyle change for TTS is reducing the activities that compress or irritate the tibial nerve.

If your job requires prolonged standing, workplace modifications help: use anti-fatigue mats, alternate between sitting and standing if possible, take walking breaks every 30 to 45 minutes to change pressure distribution, and wear supportive shoes rather than going barefoot on hard floors.

Exercise modifications matter too. If you run, reduce mileage, avoid hills and uneven terrain, and run on softer surfaces. Cross-training options such as swimming, cycling, or upper-body work let you stay active while unloading the foot, and it helps to reduce high-impact activities, jumping, and prolonged walking until symptoms settle.

In daily activities, avoid standing in one place for long periods, elevate your legs when resting, use a footrest under your desk to vary ankle position, and avoid sitting with your legs crossed for long stretches.

10.2 Footwear Strategy

Your shoes are your primary interface with the ground, so they matter a great deal here. Look for shoes with firm heel counters and medial arch support, and consider motion-control or stability running shoes if you overpronate. Replace worn-out shoes promptly, and know that over-the-counter arch supports are a reasonable, inexpensive first step before custom devices.

On the other side, avoid flip-flops, ballet flats, and unsupportive sandals, high heels, worn-out athletic shoes, and going barefoot on hard surfaces.

10.3 Sleep

Sleep and pain influence each other in both directions: poor sleep is associated with worse pain, and pain disrupts sleep. A review of this relationship concluded that the evidence for sleep disturbance predicting later pain is at least as strong as the reverse, which is a good reason to treat sleep as part of the plan rather than an afterthought.⁴²

Helpful strategies include maintaining a consistent sleep schedule, keeping your room cool (roughly 65 to 68 degrees F, or 18 to 20 degrees C), and limiting screens in the hour before bed. Elevating your legs slightly on a pillow can reduce nighttime swelling, and you might consider a night splint to keep the ankle in a neutral position (see section 8.3).

10.4 Stress Management

Chronic stress affects pain perception and makes coping harder.

Effective techniques include diaphragmatic breathing for five minutes, twice daily, and progressive muscle relaxation, which means systematically tensing and releasing muscle groups. Gentle yoga or tai chi can promote relaxation, circulation, and body awareness. Mindfulness meditation is another option worth considering: a randomized clinical trial in adults with chronic **low back pain** compared mindfulness-based stress reduction, cognitive behavioral therapy, and usual care, and found that both mindfulness-based stress reduction and cognitive behavioral therapy improved back pain and functional limitations more than usual care.⁴³ This was a back pain trial, not a foot trial, so consider it support for mindfulness as a general pain-coping skill rather than a treatment for tarsal tunnel syndrome.

10.5 Weight Management

Excess body weight increases mechanical load on the feet and can contribute to venous congestion and swelling. Higher body weight is commonly discussed as a contributing factor in foot problems generally.

A sensible approach focuses on the general eating pattern outlined above, aims for gradual and sustainable change, favors low-impact exercise such as swimming, cycling, or upper-body work, and involves your provider or a dietitian if weight loss is a goal.

10.6 Smoking Cessation

Smoking impairs peripheral circulation and oxygen delivery to tissue, and stopping is one of the highest-value health changes available to anyone with a nerve or wound healing problem. I am not attaching a trial citation to that here, because I could not verify one specific to neuropathy outcomes. The recommendation stands on well-established general health grounds, and your provider can help you access support.

11. The In-Clinic Pre-Treatment Protocol

Understand what this is for, because the distinction matters.

None of this rebuilds tissue. The loading and strengthening work in the previous chapters does that. What this protocol does is reduce the muscle tone, stiffness and tissue sensitivity that make the strength work harder to tolerate and less pleasant to do, particularly in the early weeks when things are at their most irritable.

It is preparation, not treatment. I use it in the clinic before the working part of a session, and I give it to patients to use at home before theirs. Patients who do it tend to tolerate the loading better and stick with it longer, and sticking with it is the actual mechanism by which recovery happens.

If you have to choose between doing this and doing your strength work, do your strength work.

Ankle work.

- **Foam rolling the calf.** Seated, from below the knee to above the ankle, working the inside and the outside. 60 to 90 seconds per leg, moderate pressure. Stay off the Achilles tendon itself.
- **Percussion massager**, speeds 1 to 2, level 3 only for intensity. Tibialis anterior, cushioned head, upper quadrant of the lower leg, 1 to 2 minutes per leg. Ankle ligaments, cushioned head below the ankle bone, inside and outside, 30 to 60 seconds per side per leg, staying off the bone, and skip it entirely if it is too painful. Peroneals, cushioned head or round ball, outer lower leg, 30 to 45 seconds per leg.
- **Firm trigger point tool.** Taut or tender areas around the ankle, inside and outside, 30 to 60 seconds per side per leg.
- **Foam balance pad.** Single-leg balance, 1 to 2 minutes each foot.

Foot work.

- **Percussion device**, round or flat tip, level 1 to 2. Calf upper and lower, 2 to 3 minutes. Tibialis anterior, 1 minute.
- **Massage ball or tennis ball.** Plantar fascia, rolling the bare foot over the ball through the arch and heel and any tender spots, 2 to 3 minutes per foot. Then the heel specifically, seated, small circles with sustained pressure, 60 to 90 seconds per foot.
- **Roller stick.** Calf, from above the Achilles to below the knee, 1 to 2 minutes per leg.
- **Slant board.** Stand at a level that gives a good calf stretch, heel down on the board, 1 to 2 minutes.
- **Firm trigger point tool.** Seated, foot crossed over the opposite knee, working tender areas, rolling back and forth.

Safety, and I mean these.

- Avoid all of this over an acutely injured area in the first 48 to 72 hours.
- Reduce pressure over anything actively inflamed.
- Skip percussion therapy entirely if you have a pacemaker or another implanted electronic device.
- Avoid aggressive pressure over varicose veins or compromised skin.
- Start light. Pain must not exceed 6 out of 10. Back off immediately for sharp or shooting pain.
- Stop and seek advice if you get numbness, tingling, or weakness.

One absolute rule before you roll or percuss a calf. Never work on a calf that is swollen, warm, or tight in one leg only, particularly after travel, surgery, or a period of immobilisation. That is a blood clot concern. It needs assessing the same day and it must never be massaged.

12. Red Flags: When to Seek Immediate Care

While TTS is rarely an emergency, certain symptoms warrant prompt medical evaluation.

Seek immediate care if you experience:

- Progressive weakness in the foot (inability to push off, toe walk, or spread the toes)
- Severe swelling, redness, and warmth around the ankle (possible infection or deep vein thrombosis)
- Fever with ankle or foot pain
- Sudden, severe pain after a specific injury
- Loss of sensation that is rapidly spreading or ascending up the leg
- Cold, pale, or blue foot (possible vascular compromise)
- Calf swelling with pain (possible deep vein thrombosis)

Contact your healthcare provider promptly if:

- Symptoms worsen after 4 to 6 weeks of conservative care
- Pain is severe enough to prevent sleep despite rest and elevation
- You develop new areas of numbness or tingling
- Symptoms persist beyond 3 to 6 months without improvement
- You have diabetes and notice new or worsening foot symptoms
- You have a history of significant ankle trauma or surgery

13. Building Your Personal Recovery Plan

Use this framework as a starting point for a conversation with your provider, not as a substitute for individualized care.

Week 1 to 2: Calm and Protect (Acute Phase)

If you and your provider decide to use any supplements, this is where documented deficiencies in B12 or vitamin D are worth checking and correcting. For movement, rest from aggravating activities, do gentle ankle circles, and begin nerve glides if tolerated. Therapies during this window include ice after activity for 15 to 20 minutes, elevating your legs when resting, an initial physical therapy evaluation, and considering a night splint. On the diet side, begin the general eating pattern described above and increase your hydration. Lifestyle changes to start now include switching to supportive footwear, modifying your work setup, and prioritizing sleep.

Week 3 to 6: Rebuild (Subacute Phase)

Movement progresses to advancing your nerve glides and beginning intrinsic foot strengthening, calf stretching, and ankle mobility work. Therapies typically include regular physical therapy, considering orthoses or a taping trial, and manual therapy as directed by your therapist. Continue the general eating pattern from the diet standpoint, and on the lifestyle side, implement your activity modifications, continue footwear changes, and manage stress.

Week 7 to 12: Strengthen and Restore (Rehabilitation Phase)

Movement now adds posterior tibialis strengthening, balance and proprioception work, and a gradual return to walking. Therapies shift to maintenance physical therapy visits and adjusting orthoses as needed; if progress stalls, this is the point to return to your physician and reassess the diagnosis and imaging rather than adding more modalities. Diet continues as before, and lifestyle work moves toward a gradual return to running or sport while maintaining footwear discipline.

Beyond 12 Weeks: Maintenance and Prevention

From this point, continue foot strengthening exercises 3 to 4 times per week, maintain your eating pattern, and wear supportive footwear consistently. Address flare-ups early with your established toolkit, and consider a surgical consultation if conservative care has genuinely been exhausted and symptoms remain disabling.

Final Thoughts

Tarsal tunnel syndrome can be a frustrating condition because the foot is constantly in use. Every step you take either helps or hinders your recovery. Unlike an injury to an arm or shoulder, you cannot simply immobilize your foot and wait. That makes a strategic, patient approach all the more important.

The encouraging part is that conservative management resolves symptoms for many patients, and surgery is reserved for a select group whose symptoms do not respond.⁴ Overpronation, tight calves, and unsupportive footwear are frequently discussed contributors, and those are things you can influence directly.

Nerve tissue recovers slowly, far more slowly than muscle or tendon. If your tibial nerve has been compressed for a long time, meaningful improvement can take months. Patience, consistency, and a willingness to modify daily habits matter more here than any single product or supplement.

Address your foot mechanics, keep your general health in good order, do your exercises consistently, and give your body time. If conservative measures do not provide adequate relief after a dedicated effort of several months, a surgical consultation may be reasonable. For most people, surgery is not the first step.

You are not alone in this, and there are practical, evidence-informed things you can do starting today.

References

All references below were individually verified against PubMed. PMIDs are provided so you can look up any of them directly at pubmed.ncbi.nlm.nih.gov.

1. Lau JT, Daniels TR. Tarsal tunnel syndrome: a review of the literature. *Foot Ankle Int.* 1999;20(3):201-209. PMID 10195301.
2. Ahmad M, Tsang K, Mackenney PJ, et al. Tarsal tunnel syndrome: a literature review. *Foot Ankle Surg.* 2012;18(3):149-152. PMID 22857954.
3. McSweeney SC, Cichero M. Tarsal tunnel syndrome: a narrative literature review. *Foot (Edinb).* 2015;25(4):244-250. PMID 26546070.
4. Vij N, Kaley HN, Robinson CL, et al. Clinical results following conservative management of tarsal tunnel syndrome compared with surgical treatment: a systematic review. *Orthop Rev (Pavia).* 2022;14(3):37539. PMID 36072502.
5. Jackson DL, Haglund BL. Tarsal tunnel syndrome in runners. *Sports Med.* 1992;13(2):146-149. PMID 1561508.
6. Kinoshita M, Okuda R, Morikawa J, et al. The dorsiflexion-eversion test for diagnosis of tarsal tunnel syndrome. *J Bone Joint Surg Am.* 2001;83(12):1835-1839. PMID 11741063.
7. Alshami AM, Babri AS, Souvlis T, et al. Biomechanical evaluation of two clinical tests for plantar heel pain: the dorsiflexion-eversion test for tarsal tunnel syndrome and the windlass test for plantar fasciitis. *Foot Ankle Int.* 2007;28(4):499-505. PMID 17475147.
8. Frey C, Kerr R. Magnetic resonance imaging and the evaluation of tarsal tunnel syndrome. *Foot Ankle.* 1993;14(3):159-164. PMID 8491431.
9. Nagaoka M, Matsuzaki H. Ultrasonography in tarsal tunnel syndrome. *J Ultrasound Med.* 2005;24(8):1035-1040. PMID 16040816.
10. de Souza Reis Soares O, Duarte ML, Brasseur JL. Tarsal tunnel syndrome: an ultrasound pictorial review. *J Ultrasound Med.* 2022;41(5):1247-1272. PMID 34342896.
11. Patel AT, Gaines K, Malamut R, et al. Usefulness of electrodiagnostic techniques in the evaluation of suspected tarsal tunnel syndrome: an evidence-based review. *Muscle Nerve.* 2005;32(2):236-240. PMID 16003732.
12. Gondring WH, Shields B, Wenger S. An outcomes analysis of surgical treatment of tarsal tunnel syndrome. *Foot Ankle Int.* 2003;24(7):545-550. PMID 12921360.
13. Sammarco GJ, Chang L. Outcome of surgical treatment of tarsal tunnel syndrome. *Foot Ankle Int.* 2003;24(2):125-131. PMID 12627619.
14. Turan I, Rivero-Melián C, Guntner P, et al. Tarsal tunnel syndrome. Outcome of surgery in longstanding cases. *Clin Orthop Relat Res.* 1997;(343):151-156. PMID 9345220.
15. Pfeiffer WH, Cracchiolo A 3rd. Clinical results after tarsal tunnel decompression. *J Bone Joint Surg Am.* 1994;76(8):1222-1230. PMID 8056802.
16. Gould JS. Recurrent tarsal tunnel syndrome. *Foot Ankle Clin.* 2014;19(3):451-467. PMID 25129355.
17. Sun Y, Lai MS, Lu CJ. Effectiveness of vitamin B12 on diabetic neuropathy: systematic review of clinical controlled trials. *Acta Neurol Taiwan.* 2005;14(2):48-54. PMID 16008162.
18. Ziegler D, Nowak H, Kempler P, et al. Treatment of symptomatic diabetic polyneuropathy with the antioxidant alpha-lipoic acid: a meta-analysis. *Diabet Med.* 2004;21(2):114-121. PMID 14984445.

19. Reljanovic M, Reichel G, Rett K, et al. Treatment of diabetic polyneuropathy with the antioxidant thioctic acid (alpha-lipoic acid): a two year multicenter randomized double-blind placebo-controlled trial (ALADIN II). *Free Radic Res.* 1999;31(3):171-179. PMID 10499773.
20. Ziegler D, Low PA, Litchy WJ, et al. Efficacy and safety of antioxidant treatment with alpha-lipoic acid over 4 years in diabetic polyneuropathy: the NATHAN 1 trial. *Diabetes Care.* 2011;34(9):2054-2060. PMID 21775755.
21. Sima AA, Calvani M, Mehra M, et al. Acetyl-L-carnitine improves pain, nerve regeneration, and vibratory perception in patients with chronic diabetic neuropathy: an analysis of two randomized placebo-controlled trials. *Diabetes Care.* 2005;28(1):89-94. PMID 15616239.
22. Onofri M, Fulgente T, Melchionda D, et al. L-acetylcarnitine as a new therapeutic approach for peripheral neuropathies with pain. *Int J Clin Pharmacol Res.* 1995;15(1):9-15. PMID 7490173.
23. Calder PC. Omega-3 fatty acids and inflammatory processes: from molecules to man. *Biochem Soc Trans.* 2017;45(5):1105-1115. PMID 28900017.
24. Paladini A, Fusco M, Cenacchi T, et al. Palmitoylethanolamide, a special food for medical purposes, in the treatment of chronic pain: a pooled data meta-analysis. *Pain Physician.* 2016;19(2):11-24. PMID 26815246.
25. Hesselink JM, Hekker TA. Therapeutic utility of palmitoylethanolamide in the treatment of neuropathic pain associated with various pathological conditions: a case series. *J Pain Res.* 2012;5:437-442. PMID 23166447.
26. Stracke H, Lindemann A, Federlin K. A benfotiamine-vitamin B combination in treatment of diabetic polyneuropathy. *Exp Clin Endocrinol Diabetes.* 1996;104(4):311-316. PMID 8886748.
27. Winkler G, Pál B, Nagybégányi E, et al. Effectiveness of different benfotiamine dosage regimens in the treatment of painful diabetic neuropathy. *Arzneimittelforschung.* 1999;49(3):220-224. PMID 10219465.
28. Daily JW, Yang M, Park S. Efficacy of turmeric extracts and curcumin for alleviating the symptoms of joint arthritis: a systematic review and meta-analysis of randomized clinical trials. *J Med Food.* 2016;19(8):717-729. PMID 27533649.
29. Kuptniratsaikul V, Dajpratham P, Taechaarpornkul W, et al. Efficacy and safety of Curcuma domestica extracts compared with ibuprofen in patients with knee osteoarthritis: a multicenter study. *Clin Interv Aging.* 2014;9:451-458. PMID 24672232.
30. Tarighat Esfanjani A, Mahdavi R, Ebrahimi Mameghani M, et al. The effects of magnesium, L-carnitine, and concurrent magnesium-L-carnitine supplementation in migraine prophylaxis. *Biol Trace Elem Res.* 2012;150(1-3):42-48. PMID 22895810.
31. Plotnikoff GA, Quigley JM. Prevalence of severe hypovitaminosis D in patients with persistent, nonspecific musculoskeletal pain. *Mayo Clin Proc.* 2003;78(12):1463-1470. PMID 14661675.
32. Pei Y, Liu H, Yang Y, et al. Biological activities and potential oral applications of N-acetylcysteine: progress and prospects. *Oxid Med Cell Longev.* 2018;2018:2835787. PMID 29849877.
33. Hernández-Camacho JD, Bernier M, López-Lluch G, et al. Coenzyme Q10 supplementation in aging and disease. *Front Physiol.* 2018;9:44. PMID 29459830.
34. Coppieters MW, Butler DS. Do 'sliders' slide and 'tensioners' tension? An analysis of neurodynamic techniques and considerations regarding their application. *Man Ther.* 2008;13(3):213-221. PMID 17398140.
35. Ellis RF, Hing WA. Neural mobilization: a systematic review of randomized controlled trials with an analysis of therapeutic efficacy. *J Man Manip Ther.* 2008;16(1):8-22. PMID 19119380.
36. McKeon PO, Hertel J, Bramble D, et al. The foot core system: a new paradigm for understanding intrinsic foot muscle function. *Br J Sports Med.* 2015;49(5):290. PMID 24659509.
37. Mulligan EP, Cook PG. Effect of plantar intrinsic muscle training on medial longitudinal arch morphology and dynamic function. *Man Ther.* 2013;18(5):425-430. PMID 23632367.

38. Landorf KB, Keenan AM, Herbert RD. Effectiveness of foot orthoses to treat plantar fasciitis: a randomized trial. *Arch Intern Med.* 2006;166(12):1305-1310. PMID 16801514.
39. Rome K, Brown CL. Randomized clinical trial into the impact of rigid foot orthoses on balance parameters in excessively pronated feet. *Clin Rehabil.* 2004;18(6):624-630. PMID 15473114.
40. Radford JA, Landorf KB, Buchbinder R, et al. Effectiveness of low-Dye taping for the short-term treatment of plantar heel pain: a randomised trial. *BMC Musculoskelet Disord.* 2006;7:64. PMID 16895612.
41. Vickers AJ, Vertosick EA, Lewith G, et al. Acupuncture for chronic pain: update of an individual patient data meta-analysis. *J Pain.* 2018;19(5):455-474. PMID 29198932.
42. Finan PH, Goodin BR, Smith MT. The association of sleep and pain: an update and a path forward. *J Pain.* 2013;14(12):1539-1552. PMID 24290442.
43. Cherkin DC, Sherman KJ, Balderson BH, et al. Effect of mindfulness-based stress reduction vs cognitive behavioral therapy or usual care on back pain and functional limitations in adults with chronic low back pain: a randomized clinical trial. *JAMA.* 2016;315(12):1240-1249. PMID 27002445.
44. Germini F, Coerezza A, Andreinetti L, et al. N-of-1 Randomized Trials of Ultra-Micronized Palmitoylethanolamide in Older Patients with Chronic Pain. *Drugs Aging.* 2017;34(12):941-952. PMID: 29210011
45. Stracke H, Gaus W, Achenbach U, Federlin K, Bretzel RG. Benfotiamine in diabetic polyneuropathy (BENDIP): results of a randomised, double blind, placebo-controlled clinical study. *Exp Clin Endocrinol Diabetes.* 2008;116(10):600-605. PMID: 18473286
46. Maroon JC, Bost JW. Omega-3 fatty acids (fish oil) as an anti-inflammatory: an alternative to nonsteroidal anti-inflammatory drugs for discogenic pain. *Surg Neurol.* 2006;65(4):326-331. PMID: 16531187

This guide was compiled from peer-reviewed medical literature that was individually verified for this edition. It is intended to give you knowledge and options to discuss with your healthcare provider. It does not replace medical care, and it does not promise a cure.

Working with Root Health

Burning or numbness in the foot deserves an examination before any program is assumed to fit. If no one has examined the nerve, I offer a free Root Discovery Session at Root Health to talk through what you are dealing with and what a reasonable next step looks like. You can schedule one at rootcauseunlocked.com/discovery, or reach the office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the free Root Cause Unlocked tracker has a mode built for musculoskeletal recovery, at rootcauseunlocked.com/tracker. Eight weeks of it turns “I think it might be helping” into something you and I can both read.

© 2026 · For personal educational use only